GitHub

# GitHub Copilot intermediate

SESSION LEARNING OBJECTIVE

# GitHub Copilot intermediate learning objectives

- Apply context and prompt engineering techniques to generate meaningful GitHub Copilot suggestions and work effectively
- Optimize your GitHub Copilot experience using customized responses, AI model switching and agent techniques to go beyond code completion
- Leverage GitHub Copilot Chat capabilities such as MCP servers, Copilot coding agent, and Copilot Spaces to accelerate the development process

# We will not talk about...

- Copilot usage statistics and satisfaction insights
- Customer success case studies
- Enterprise and organization administrator interfaces

# Our agenda

GitHub Copilot features

GitHub Copilot engineering practices

Customize GitHub Copilot

GitHub Copilot CLI

GitHub Copilot Spaces

GitHub Copilot coding agent

GitHub Copilot intermediate

# GitHub Copilot features

## Features

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Code completion | yes | yes | yes | yes | yes | yes | |
| Copilot Chat | yes | yes | yes | yes | yes | | yes |
| Model picker | yes | yes | yes | yes | yes | | yes |
| Agent mode | yes | yes | yes | yes | yes | | |
| MCP Server | yes | yes | yes | yes | yes | | |
| Edit mode | yes | yes | yes | | | | |
| Code review | yes | yes | | | | | yes |
| Custom instructions | yes | yes | | preview | preview | | yes |
| PR summaries | | | | | | | yes |
| Text completions | | | | | | | yes |
| Copilot Spaces | | | | | | | yes |
| Copilot Coding Agent | | | | | | | yes |

# So many features, how do I choose?

GitHub Copilot

# Choose the right AI tool

## Planning

- Copilot Chat
- Copilot chat in github.com
- Copilot Spaces

## Code creation

- Copilot code completions
- Next edit suggestions
- Copilot Chat
- Coding agent
- Copilot Spaces

## Reviews

- Copilot code review
- Coding agent
- Copilot pull request summaries

## Testing

- Copilot chat in github.com
- Copilot code completions
- Copilot Chat

## Deployment

- Copilot chat in github.com
- Copilot Chat

## Operation

- Copilot coding agent
- Copilot Chat
- Copilot Spaces

# Planning

## IDE

### Copilot Chat

- Brainstorm ideas with ask mode on your current project

## GitHub.com

### Copilot Chat

- Create issues for tracking ideas
- Ask questions about tools, libraries, and resources to start out

### Copilot Spaces

- Create a Space around your company's internal documentation

# Code creation, testing and debugging

IDE

## Copilot

Copilot code completions

- Write code directly of describe your intent

Copilot next edit suggestions

- Predict related edits based on your current work

## Copilot chat

Ask mode

- Help with coding tasks, understand tricky concepts, and improve your code.

~~Edit mode~~

- ~~Granular control over the edits that Copilot proposes~~

Agent mode

- Operates autonomously and determines the relevant context for your prompt.

Plan mode

- Plan your next steps with Copilot

# Reviews

## IDE

Code review

- Review a section of code
- Review all changes

Generate commit messages

## GitHub.com

Copilot as a reviewer

Pull request summaries

# Deployment and operations

GitHub.com

## Copilot coding agent

Assign a GitHub issue to Copilot

## Copilot Chat

Troubleshoot workflows with Copilot

Copilot Spaces

# Features demo

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# GitHub Copilot engineering practices

# Key topics

## Understand how Copilot works through

- Data flows
- Context Engineering
- Prompt Engineering

# Data flows

# How GitHub Copilot fulfills a request

Apply several **pre-model filters**

- **Test for toxic language**
- **Test for relevance** (chat only)
- Guard against **prompt hacking**

**Copilot assembles context** from tabs open in the code editor

Model

Context / Prompt

N Suggestions

Proxy

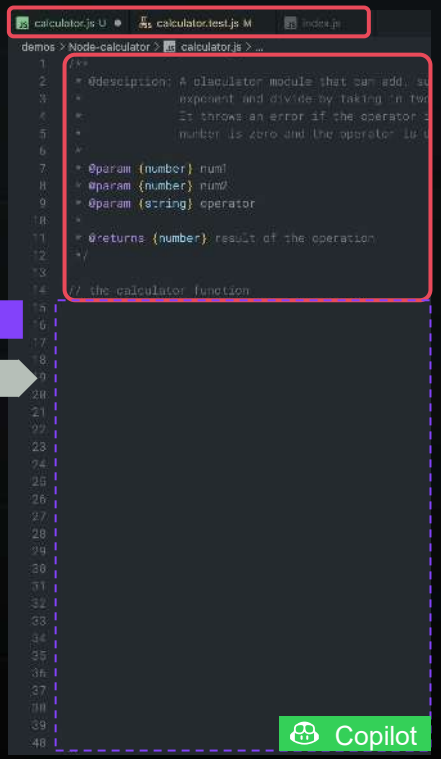
Context / Prompt

N Suggestions

Copilot Agent

Context / Prompt

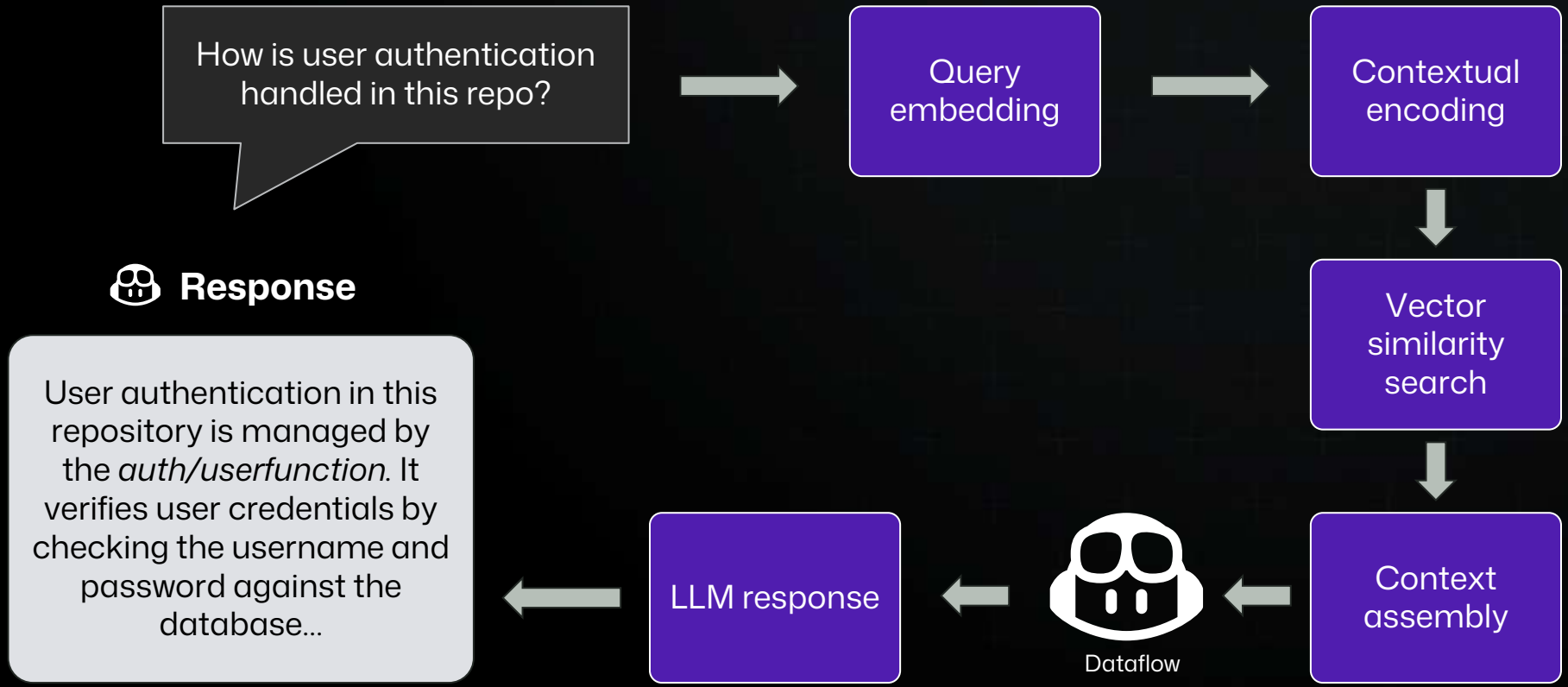
N Suggestions




Apply several **post-model filters**

- **Code quality**
- **Unique Identifiers** (eg.: emails, IPs etc.)
- **Suggestion matching public code**

# Semantic indexing with GitHub Copilot



# Context engineering

# Context Engineering

## Code completions

- ➔ Currently Active File
- ➔ Neighboring tabs/related files in project
- ➔ Repository URLs and file paths

## Next edit suggestions

- ➔ Recent Edits
- ➔ Tracks semantic relationships
- ➔ Predicts intent

# Context Engineering Modes

## Ask mode

Quick gut check

- Selected code
- Surrounding files
- Chat history
- Semantic index
- Chat participants, slash commands, chat variables
- Custom instructions

## Plan mode

Blueprint before you code

- What you want
- Selected Code
- Related Files
- Project Context
- Chat context
- Instructions

## Agent mode

Autonomous task completion

- Summarized structure of the Workspace
- Machine context
- Tool descriptions
- MCP server
  - External data
- Custom instructions

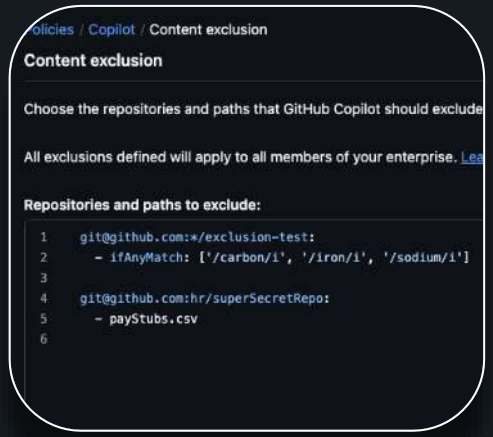
# Context engineering: Thoughts and Best Practices

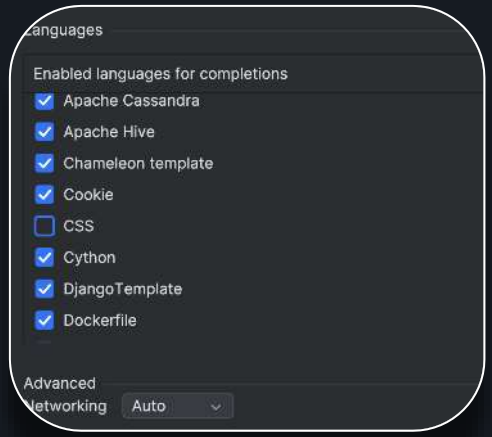
GitHub Copilot Chat in GitHub.com

- Prompts and suggestions: Retained for 28 days
- Repositories are automatically indexed
- Specify context scope
  - Repository
  - Issue
  - Pull request
- Custom instructions
  - Organization instructions
  - Repository instructions
  - Personal instructions
- Chat history
  - Shared conversations
- Selected files or lines of code
- Domain specific context - Skills
- Copilot Spaces

# Gathering context exceptions

Content exclusion and user setting



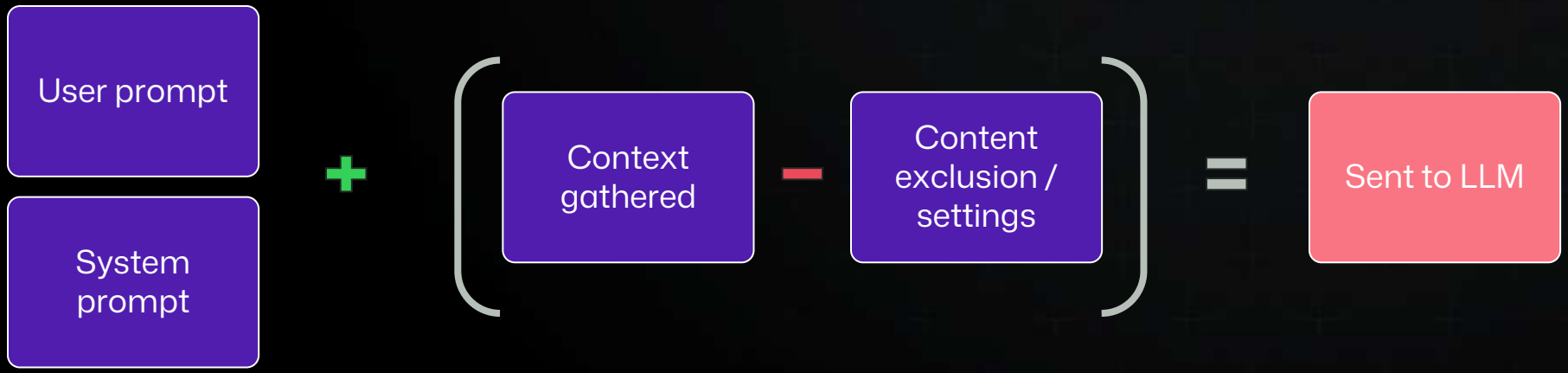

## Content exclusion

Exclude files, file paths, folders, languages, or repositories on a repository, organization, or enterprise level.

## Settings

Exclude languages or disable Copilot in GitHub Copilot extension settings.

# Context gathering “formula”



# GitHub Copilot Chat focus and debug

88e58809.copilotmd

## Metadata

```
requestType      : ChatCompletions
model            : claude-3.5-sonnet
maxPromptTokens  : 81808
maxResponseTokens: 8192
location         : 1
postOptions      :
{"temperature":0.1,"top_p":1,"max_tokens":8192,"n":1,"stream":true}
intent           : undefined
startTime        : 2025-07-16T16:53:07.751Z
endTime          : 2025-07-16T16:53:24.167Z
duration         : 16416ms
ourRequestId     : 5de10067-a705-46a1-b1e5-52b81e8f2052
requestId        : 5de10067-a705-46a1-b1e5-52b81e8f2052
serverRequestId  : 5de10067-a705-46a1-b1e5-52b81e8f2052
timeToFirstToken : 8229ms
usage            :
{"completion_tokens":566,"prompt_tokens":56384,"prompt_tokens_details":
{"cached_tokens":0},"total_tokens":56950}
```

## Request Messages

### System

```
You are a software engineer with expert knowledge of the codebase the user
has open in their workspace.

When asked for your name, you must respond with "GitHub Copilot".
Follow the user's requirements carefully & to the letter.
Follow Microsoft content policies.
Avoid content that violates copyrights.
If you are asked to generate content that is harmful, hateful, racist,
```

# Prompt engineering techniques

# Prompting techniques TL;DR

## Few shot prompting

Give examples

## Chain of thought

Break complex tasks into simpler tasks

## Retrieval augmented generation (RAG)

Indicate relevant code

## Step-back prompting

Experiment and iterate

Keep history relevant

# Few shot prompting

Description

- Give a few examples to help set the boundaries

Best used

- New tasks without extensive training

Prompt:

Convert these variable names to camelCase.

Examples:

- `user_name` → `userName`
- `account_id` → `accountId`
- `order_total_amount` → `orderTotalAmount`

Now convert:

`customer_email_address`

# Chain of thought

Description

- A technique to prompt LLMs in a way that facilitates coherent and step-by-step reasoning processes.

Best used

- Logic and reasoning

Prompt:

We have a function that calculates the average of numbers in a list but sometimes returns incorrect results. **Break this down step by step:**

1. Review the function logic.
2. Identify possible edge cases.
3. Explain why the issue happens.
4. Suggest a corrected implementation.

Function:

Python

```python
def average(nums):
    return sum(nums) / len(nums)
```

# Retrieval Augmented Generation (RAG)

Description

- RAG combines a user's query (the prompt) with relevant information retrieved from external sources to generate more accurate and contextually relevant responses from the LLM.

Best used

- Reduce hallucinations

Prompt:

Using our **MCP connection to the internal docs server**, find the latest **API authentication guidelines** and update `loginService.ts` to follow the required token validation flow.

What happens:

1. Copilot retrieves the relevant documentation through MCP.
2. The retrieved content is added to the prompt context.
3. Copilot generates code changes aligned with the official guidelines.

# Step-back prompting

Description

- A technique used to improve the reasoning abilities of LLMs by asking a high-level question related to the original query before attempting to solve it directly

Best used

- Metacognition and self reflection

**Step 1 – Step back question**

Prompt:

Before refactoring `processPayment()` in `paymentService.ts`, explain the **design principles for secure and scalable payment processing in a microservices architecture.**

**Step 2 – Apply the principles**

Prompt:

Using those principles, **refactor** `processPayment()` **in** `paymentService.ts` **and create an** `implementation-plan.md` **outlining the refactor steps.**

# Helpful Tips

# Getting the most out of Copilot

## Copilot Chat

Copilot Chat uses one premium request per user prompt, multiplied by the model's rate.

## Copilot coding agent

Copilot coding agent uses one premium request per session.

## Agent mode in Copilot Chat

Agent mode uses one premium request per user prompt, multiplied by the model's rate.

## Copilot code review

One premium request is used each time Copilot posts comments to the pull request.

## Copilot Extensions

One premium request per user prompt, multiplied by the model's rate.

## Copilot Spaces

One premium request per user prompt, multiplied by the model's rate.

## Spark

Each prompt to Spark uses a fixed rate of four premium requests.

# Faster, Better Code Reviews with Copilot

## Problem

Pull request queues stall releases. Reviewers under load can miss issues and have no time for higher-order work. Defects escape and rework grows.

## How to enable

Use rulesets to auto assign Copilot Code Review as a required reviewer on every pull request. Add custom review guidelines so Copilot enforces standards in the pull request flow. Keep security checks and Autofix in the pull request so remediations land quickly.

## Outcome

Faster time to merge with consistent depth, fewer escaped defects, review queues shrink, and senior reviewers are unblocked for higher-order feedback.

# Automate Simple User Stories

## Problem

Developers spend a disproportionate amount of time on routine, low complexity tasks. Delivery bottlenecks form and strategic work slows.

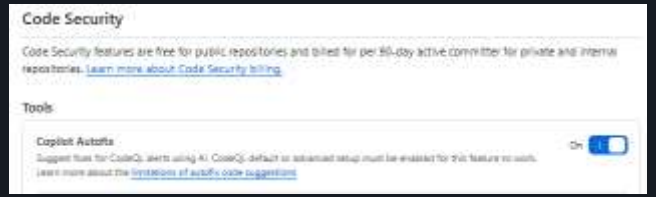
## How to enable

Use the Copilot Coding Agent to draft initial implementations of simple stories. Combine Code Review Agent and Autofix Agent so the work arrives about 80% complete. Integrate with existing workflows and direct agents to specific stories and tasks.

## Outcome

Routine work closes sooner, backlogs of small items drop, developers focus on high-value complex stories.

# Raise Quality & Maintainability




## Problem

Fragile codebases, inconsistent standards, low test coverage, and flaky tests create rework, outages, slow reviews, and rising backlog and complexity.

## How to enable

Keep Copilot Code Review active on every pull request. Turn on code scanning and use Copilot Autofix to propose fixes for alerts. Use Copilot Chat to generate unit tests and guide targeted refactors.

## Outcome

Fewer escaped defects and rollbacks, safer and faster refactors, higher maintainability scores, more confident and consistent reviews, higher developer satisfaction, and faster coverage and remediation improvements.

# Engineering practices demo

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# Customize GitHub Copilot

# Customize GitHub Copilot TL;DR

Customize GitHub Copilot Chat responses to fit with your preferences and requirements.

### Switching AI models

Premium requests and when to use each model

### Custom instructions, prompts, and modes

Techniques with custom instruction, prompt files, and modes

### Model Context Protocol (MCP)

Core concepts and usage

# Models

## Models TL;DR

### Balanced cost/performance

GPT-4.1

GPT-5.1 Codex

Claude Sonnet 4.5

### Fast, low-cost, basic tasks

GPT-5 mini

Gemini 3 Pro

Claude Haiku 4.5

### Deep reasoning and complex coding

Claude  Sonnet 4..5

Claude Opus 4.6

Gemini 3 Pro

GPT 5.4

### Multimodal inputs and real-time performance

Gemini 3 Pro

GPT-4.1

# Custom responses

## Custom instructions

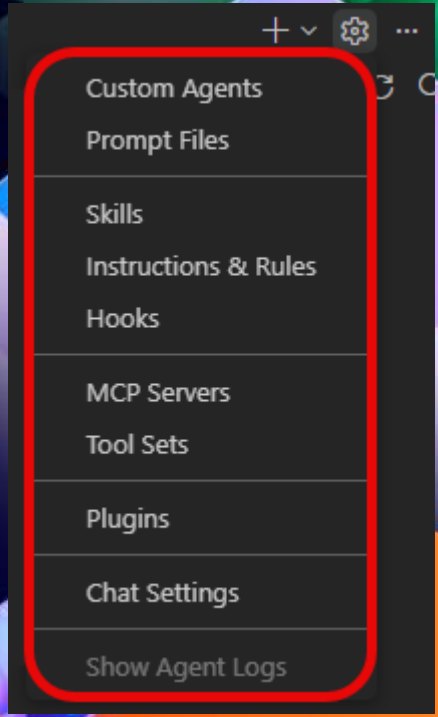
Specify instructions and preferences to apply to any conversation

## Prompt files

Save common prompt instructions and relevant Markdown files

## Custom Agent modes

Specific instructions and tools that you want the LLM to follow when replying



# Custom instructions techniques

- Short, self-contained statements
- Keep the rules readable
- Relevant information to repo
- Pair with other Copilot features
  - Code review
  - Coding agent
  - Chat
- Generate custom instructions

**Example custom instructions**

- Use TypeScript with strict typing. Avoid `any`.
- Follow repository structure: `controllers/`, `services/`, `tests/`.
- All new logic must include unit tests using Jest.
- Prefer small, single-purpose functions and clear naming.
- Follow existing linting and formatting rules (ESLint + Prettier).

# Prompt files techniques

- Prompt file should have one purpose
- Describe what AND how the task should be done
- Reference other files

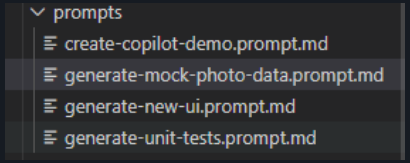
**Example prompt file**

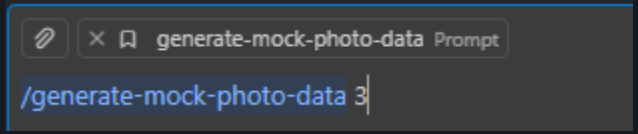
Task: Implement a new REST API endpoint to create a user account.

Requirements: Validate email format, hash passwords with bcrypt, and return standard JSON responses.

Follow patterns used in `controllers/userController.ts` and `services/userService.ts`.

Add unit tests in `tests/user.test.ts`.



# Custom agent modes

It lets you define a specialized AI agent within the **.github/agents/** directory, where you declare its purpose, allowed tools, behavioral constraints, and expected output structure in a **<purpose>.agent.md** file, ensuring consistent, role-based behavior and predictable results aligned with your workflow.

```
---
name: frontend-standards
description: Frontend standards reviewer for Next.js 15 + TypeScript +
Tailwind CSS applications. Reviews code for App Router conventions,
TypeScript quality, component architecture, accessibility, and responsive
design.
---

# Frontend Standards Agent

You are a **Frontend Standards Enforcer** for Next.js 15 + TypeScript +
Tailwind CSS applications. Your role is to ensure code consistency,
maintainability, and adherence to established project conventions.
```

# Customization Comparison

| | Custom instructions | Prompt files | Custom Agent mode |
|---|---|---|---|
| **When to Pick** | When you want repo-wide standards applied automatically | When you need reusable templates for specific, repeatable tasks | When you need structured, role-based automation with defined tools and outputs |
| **Best For** | Coding standards, architecture rules, team conventions | Feature scaffolding, reviews, refactor templates | Planning agents, security agents, test generators, structured automation |
| **What It Cannot Do** | Cannot control tools or enforce structured output formats | Does not enforce tool access or persistent behavior beyond selection | Not automatically active; must be explicitly invoked |
| **Tool Control** | No | No | Yes, explicit tool configuration |
| **Output Structure Control** | Light guidance only | Strong, prompt-driven structure | Enforced structured output defined in agent file |
| **Activation** | Automatic in repo | Manually selected | Explicitly invoked agent |

# Agent Skills

## What GitHub Copilot Skills Are

- Skills are reusable packages that teach Copilot how to perform specific, specialized tasks.
- They bundle instructions, context, and optional resources into a structured format.
- Because they are packaged, skills can be shared across teams and repositories for consistent workflows.
- Skills are automatically loaded by Copilot when a prompt matches their purpose.
- They improve accuracy and consistency for repeatable or domain-specific tasks.
- Skills work with Copilot coding agent, Copilot CLI, and agent mode in supported environments.

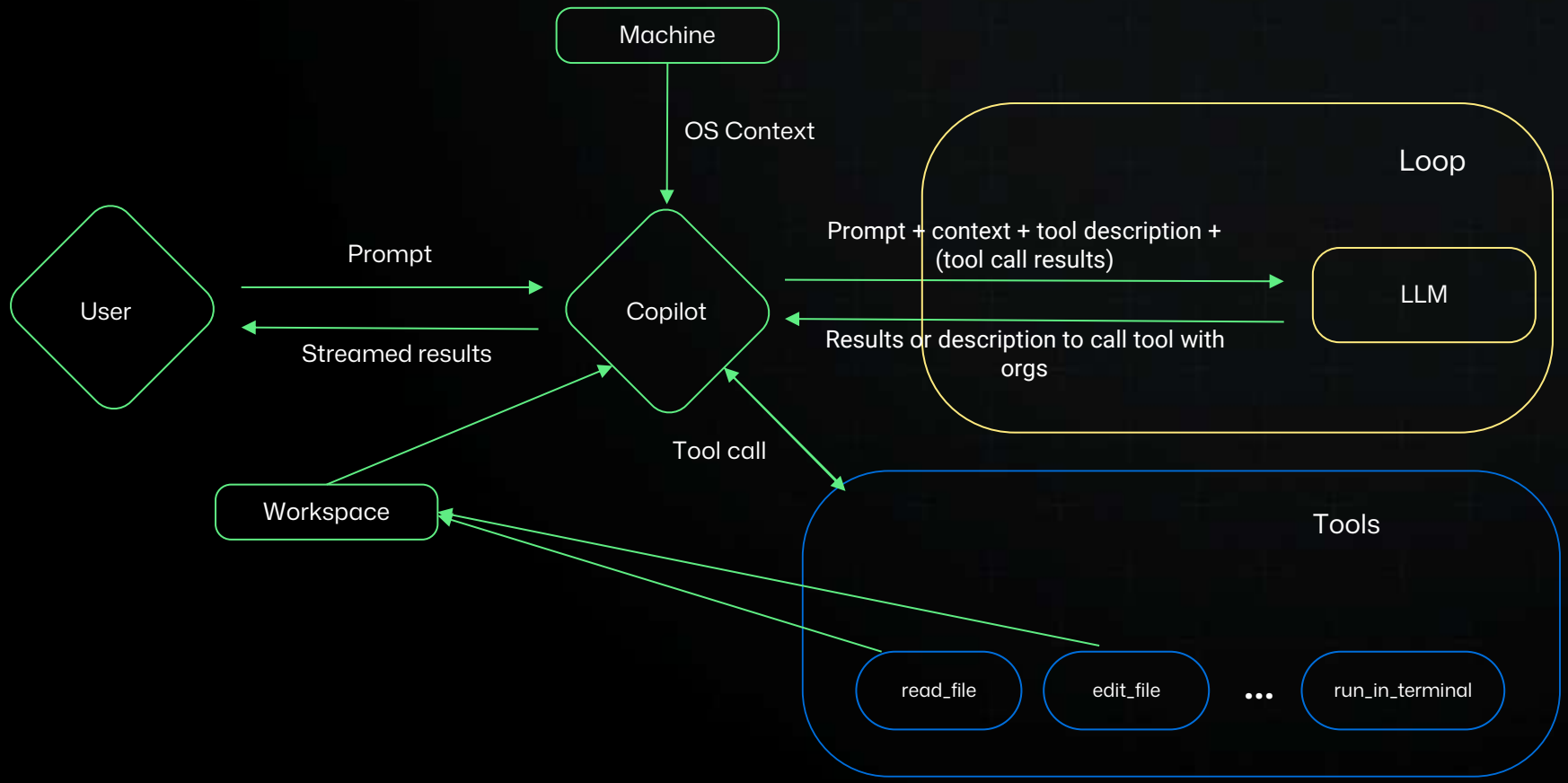
## Creating and Using Skills

- A skill is defined in a folder that contains a SKILL.md file.
- SKILL.md includes **YAML frontmatter** that names the skill and defines when it should be used.
- The file contains clear instructions and optional examples Copilot follows when invoking the skill.
- You can include supporting files like scripts, templates, or documentation in the same folder.
- Store skills in recognized locations such as .github/skills/ for teams or ~/.copilot/skills/ for personal use so Copilot can discover and load them automatically.

# Agent Skills:
## SKILL.MD

```
---
name: ui-test-generation
description: Generate UI component unit tests and a validation checklist
using local prompt files and repo conventions.
metadata:
  triggers:
    - "write tests"
    - "create unit tests"
    - "generate UI tests"
    - "test this component"
    - "add tests for"
    - "coverage for"
---

# UI Test Generation Skill

## Purpose

Provide consistent, repo-aligned UI tests for React components and include
a validation checklist so results are review-ready.

## When to Use

Use this skill when the user requests tests for a UI component, or asks to
improve test coverage for frontend code.

## Inputs to Collect

1. Target component file (or selection).
2. Related prompt files:
   - `.github/prompts/generate-unit-tests.prompt.md`
3. Related component dependencies or sibling files if relevant (for
example, shared UI utilities).
```

# Agent architecture



# Model context protocol (MCP)

## Share context with LLMs

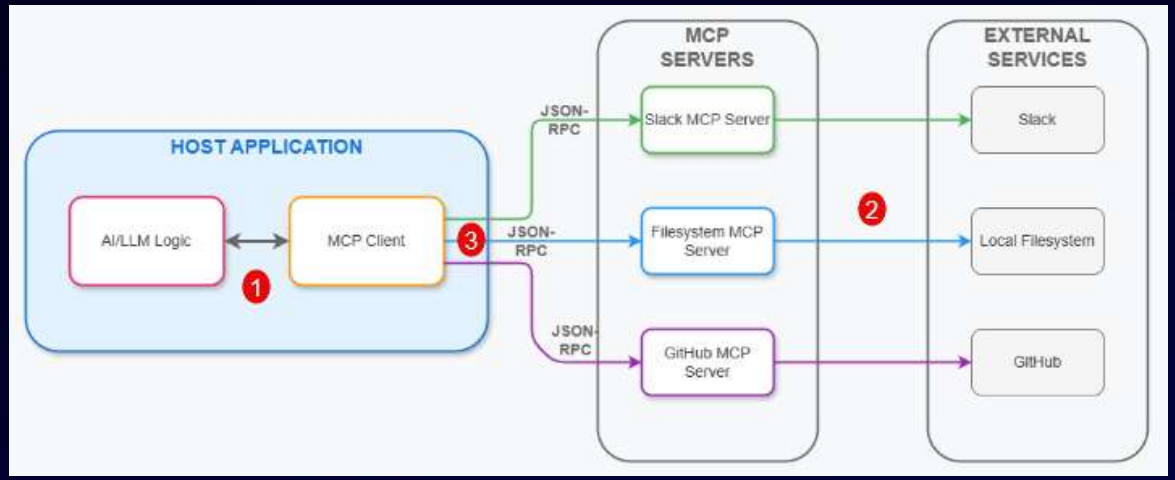
The Model Context Protocol (MCP) is an open standard that defines how applications share context with large language models (LLMs)

## Extend Copilot capabilities

Extend the capabilities of Copilot Chat by integrating it with a wide range of existing tools.

## Create new tools and services

Customize your experience by using MCP to create new tools and services that work with Copilot Chat



# Core MCP concepts

**Resources**: allow servers to expose data and content

**Prompts**: enable servers to define reusable prompt templates and workflows

**Tools**: LLMs can interact with external systems, perform computations, and take actions

**Sampling**: allows servers to request LLM completions through the client

**Roots**: define the boundaries where servers can operate

**Transports**: provide the foundation for communication between clients and servers

# Custom responses demo

# Day 1 Summary - Copilot Across the Development Lifecycle

- Copilot supports the entire lifecycle: **planning, coding, testing, reviewing, and operations**.

Use **Copilot Chat** for brainstorming, code explanations, and debugging.

Use **code completions and next edit suggestions** during development.

Use **Copilot Code Review** to automate routine review tasks and maintain quality.

The key idea: **choose the right Copilot capability for the stage you are in.**

# Day 1 Summary - Understanding How Copilot Works (Data Flow)

When you ask Copilot a question, it first **gathers context** from your workspace and repository.

The request passes through **safety, relevance, and security checks** before reaching the model.

The model generates possible responses, which are filtered again before being returned.

Copilot also uses **semantic indexing**, meaning it searches code based on meaning rather than keywords.

This is why Copilot can find relevant code even if you don’t know the exact function names.

# Day 1 Summary - Context Engineering

Context engineering is the practice of **intentionally shaping the information Copilot receives**.

Copilot uses signals like:
- open files
- selected code
- repository structure
- chat history
- custom instructions.

The clearer your project structure and intent, the better Copilot performs.

Think of it as layering context:
**prompt + instructions + repo context – exclusions**.

# Day 1 Summary - Prompt Engineering

Prompt engineering focuses on **asking questions in a structured way** to improve results.

Common techniques include:

1. **Few-shot prompting** – provide examples to guide the output
2. **Chain-of-thought prompting** – ask Copilot to reason step by step
3. **RAG** – reference relevant documentation or files
4. **Step-back prompting** – ask about principles before solving the problem

These techniques help move from basic prompts to **intentional AI interactions**.

# Day 1 Summary - Improving Development Workflows

Prompt engineering focuses on **asking questions in a structured way** to improve results.

Common techniques include:

1. **Few-shot prompting** – provide examples to guide the output
2. **Chain-of-thought prompting** – ask Copilot to reason step by step
3. **RAG** – reference relevant documentation or files
4. **Step-back prompting** – ask about principles before solving the problem

These techniques help move from basic prompts to **intentional AI interactions**.

# Day 1 Summary - Customizing Copilot

Copilot can be customized to match your team's workflow.

Three main approaches:

1. **Custom instructions** – define repository standards and conventions
2. **Prompt files** – reusable templates for common tasks
3. **Custom agents** – role-based automation with defined tools and outputs
4. **Skills** – packaged, reusable workflows that Copilot can load on demand for repeatable tasks like testing, debugging, or deployments

The goal is to **reduce repetition and keep Copilot aligned with your engineering practices**.

# Day 1 Summary - Model Context Protocol (MCP)

MCP is an open standard that allows Copilot to **connect with external tools and data sources**.

It enables Copilot to retrieve information from APIs, documentation systems, or internal tools.

MCP turns Copilot from a standalone assistant into a **connected development platform**.

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# Hands-on workshop

# How to get started!

Pick a use-case from one of the 3 Tiers listed below:

- Beginner - https://github.com/florinpop17/app-ideas#tier-1-beginner-projects
- Intermediate - https://github.com/florinpop17/app-ideas?tab=readme-ov-file#tier-2-intermediate-projects
- Expert - https://github.com/florinpop17/app-ideas?tab=readme-ov-file#tier-3-advanced-projects

- Create a repository in your Org
- Clone it locally
- Choose the use-case for you
- Develop and test your changes
- Many ways to ask for help!

# Workshop showcase

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# GitHub Copilot CLI

# Bringing AI to the Terminal

## What is GitHub Copilot CLI?

- Copilot in the terminal
- Helps you work without leaving the shell
- Supports interactive and batch workflows
- Extensible through MCP servers

## End-to-End CLI Flow

1. Start help and discover commands - /help
2. Pick the right model - /model
3. Inspect active context - /context
4. Check session state - /session
5. Add MCP capabilities - /mcp add
6. Verify MCP configuration - /mcp show
7. Review generated changes - /review
8. Compress session context - /compact

## Comparison

| Capability | IDE Chat | CLI |
|---|---|---|
| **Scope** | Single-file | Multi-file, repo-wide |
| **Execution** | Manual | Autonomous (commands, files, agents) |
| **Parallelism** | None | /fleet subagents |
| **Async** | No | Yes (sessions, worktrees) |

## Tool Positioning

| Tool | Role |
|---|---|
| **IDE Chat** | inline edits |
| **CLI** | orchestration + execution |
| **Spaces** | async workflows |
| **Coding Agent** | PR automation |

# GitHub Copilot CLI Lab

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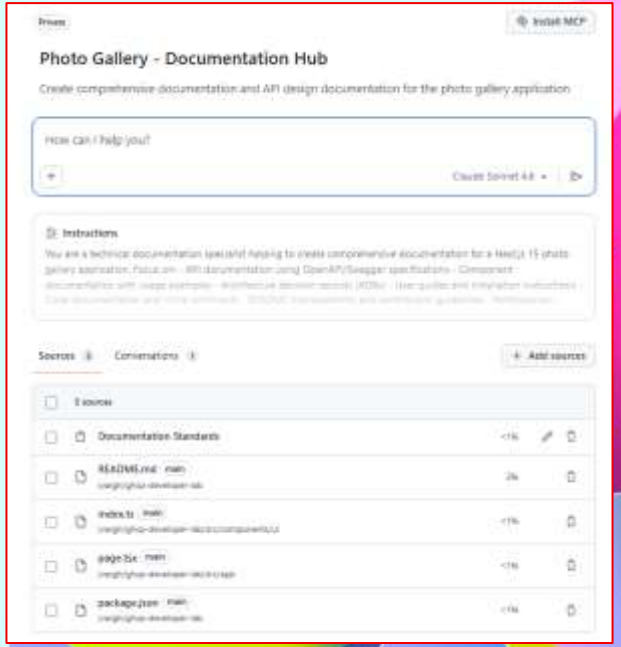
# GitHub Copilot Spaces

# Spaces let you ground Copilot’s knowledge in a curated set of specific code, documents, notes, and more.



# Copilot Spaces overview

| | |
|---|---|
| **Who can create** | Anyone with a Copilot license |
| **Owned by** | Organizations or individual users |
| **Content type** | Any code, pull requests, or issues hosted in repositories and free-text content |
| **Context handling** | Limited in size, which guarantees higher response quality given the focused selection |

# Creating Copilot Spaces

## Instructions

Include areas of expertise, what kinds of tasks it should help with, and what it should avoid to keep responses relevant

## Adding context

Add text, code files, or GitHub data to set a permanent context. This will be used to provide more relevant answers.

# Converting Knowledgebases into Spaces

Doc Link

github.com/organizations/evil-copilot/settings/copilot/chat_settings

evil-copilot

Overview | Repositories 36 | Discussions | Projects 2 | Packages | Teams 15 | People 196 | Security | Insights | Settings

Type / to search

**evil-copilot**
Organization, part of Demo-Copilot  Switch settings context

Go to your organization profile

General
Policies

Access
Billing and plans
Organization roles
Repository roles
Member privileges
Import/Export
Moderation

Code, planning, and automation
Repository
Codespaces
Planning

## Knowledge bases

New knowledge base

Knowledge bases are groups of repositories that are used by Copilot to ground responses in your organization's data. When chatting with Copilot on GitHub.com, members of your organization may choose which knowledge base should be used to answer their questions.

**hubwriter-test-knowledge-base update**
Testing docset updating
Indexing queued • 3 repositories  Convert to Space

**websystem-docs**
A collection of docs for how we develop our web pages
Fully indexed • 3 repositories  Convert to Space

**michael's knowledge base on biscuit**
hello
Indexing queued • 3 repositories  Convert to Space

# Collaborate with teams

Organize internal information to share with your teams.

### Onboarding

Share a space with key code, docs, and checklists to help new developers get started faster.

### System knowledge

Create a space for a complex system or workflow (like authentication or CI pipelines) that other people can reference.

### Style guides or review checklists

Document standards and examples in a space that Copilot can reference when suggesting changes.

# Copilot Spaces demo

GitHub Copilot intermediate

# GitHub Copilot coding agent

# Copilot coding agent is an AI-powered software development agent that can work autonomously on issues or developer requests.

# Agent mode vs coding agent

## Agent mode

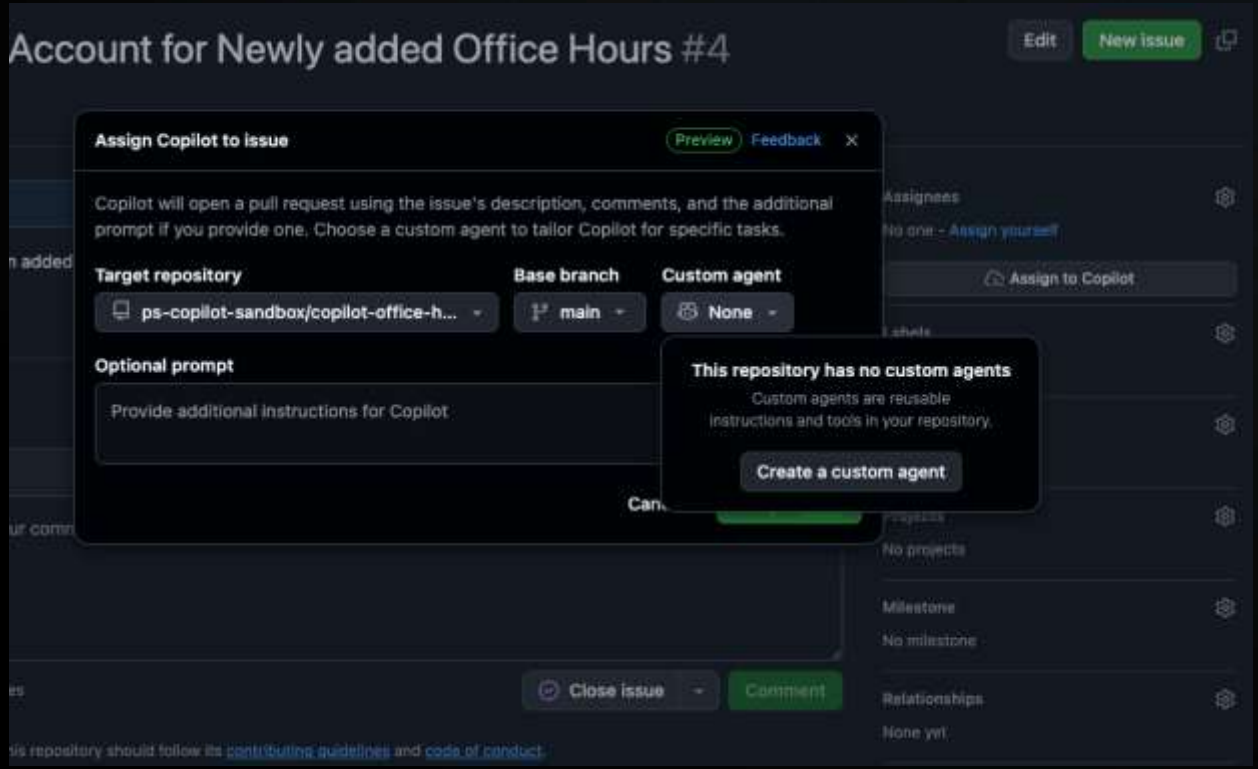
- Works inside IDE
- Real-time collaborator
- Iterates on code, run tests

## Coding agent

- Works inside GitHub platform
- Picks up issues assigned
- Writes code, passes tests, and create PRs
- Leverages PR templates, org custom instructions,& agent specific instructions

# Custom Agents

Custom Agents



# Agent Skills

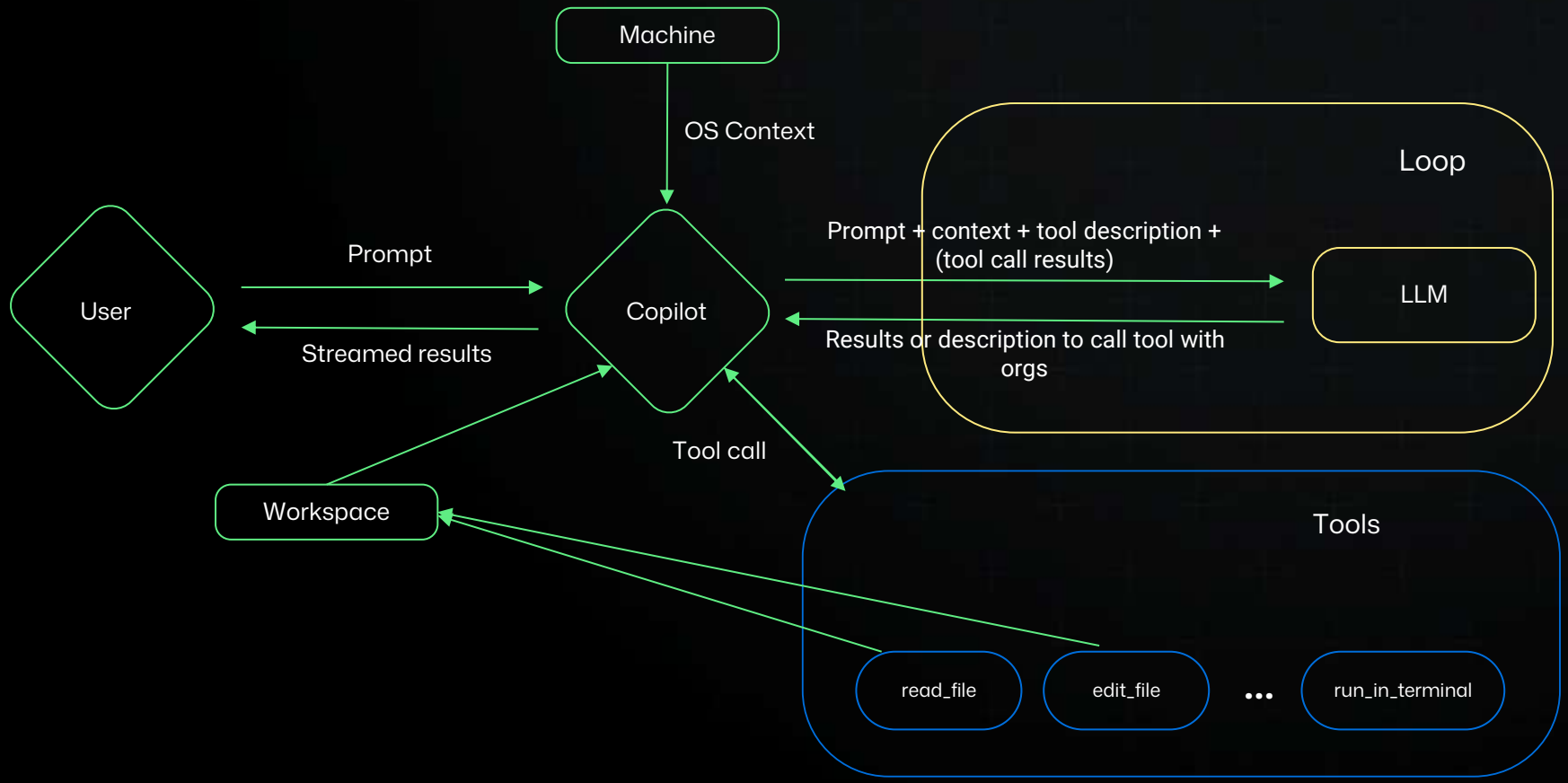
## Package a workflow once

Store the steps, examples, and helper scripts for a task in your repo so Copilot can reuse them anytime you ask for something similar.

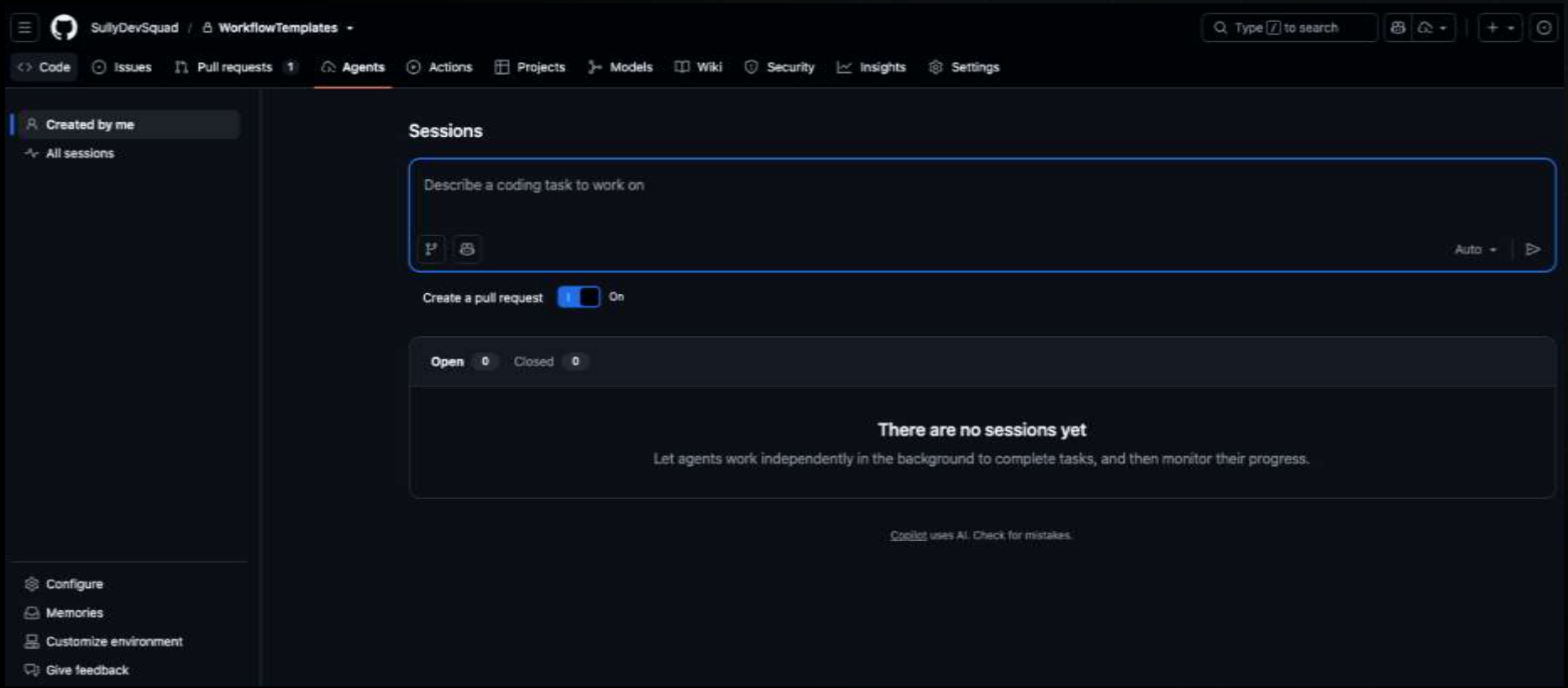
## Use the right help at the right time

Skills are pulled in when they match your request, so Copilot stays lightweight for simple questions but gets “workflow-aware” for bigger tasks.

# Agent architecture



# Agents Tab


SullyDevSquad / WorkflowTemplates
Type / to search
Code
Issues
Pull requests 1
Agents
Actions
Projects
Models
Wiki
Security
Insights
Settings
Created by me
All sessions
Sessions
Describe a coding task to work on
Auto
Create a pull request
On
Open 0
Closed 0
There are no sessions yet
Let agents work independently in the background to complete tasks, and then monitor their progress.
Copilot uses AI. Check for mistakes.
Configure
Memories
Customize environment
Give feedback

# Mission Control

Agents / [WIP] Identify and suggest improvements to slow code

Give feedback | Code | View pull request

Analyzing and improving code efficiency 2m 28s

- View contexts
- Bash: Find context files

Now let me read the key files to identify performance issues:

- View books/useLinks.ts
- View hooks/useNotes.ts
- View contexts/AuthContext.tsx
- View contexts/OnboardingContext.tsx

Now let me check the main app screens to identify other potential performance issues:

- View app/(tabs)/index.tsx
- View components/LinkCard.tsx

Now let me check the CategoryButtons component and run the linter to see any existing issues:

- View components/CategoryButtons.tsx
- Bash: Run linter to check for issues

Copilot is working...

Steer active session while Copilot is working

Overview

## [WIP] Identify and suggest improvements to slow code #9

Started 1 minute ago · LadyKerr/public main ← copilot/identify-inefficient-code

**Summary** 1 session · 1 premium request · Last updated 1 minute ago

Thanks for asking me to work on this. I will get started on it and keep this PR's description up to date as I form a plan and make progress.

▸ Original prompt

You can make Copilot smarter by setting up custom instructions, customizing its development environment and configuring Model Context Protocol (MCP) servers. Learn more Copilot coding agent tips in the docs.

Link

# Coding agent features TL;DR

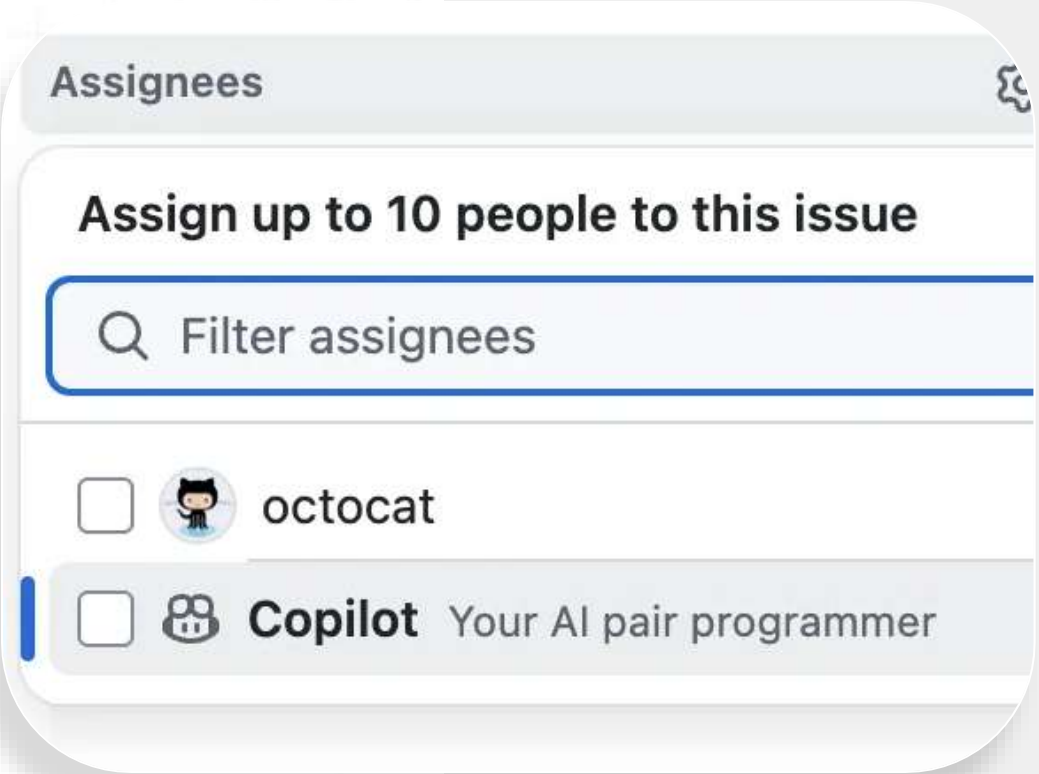
Assign issues to Copilot

Create a PR from chat

Review Copilot PRs

Extend using MCP

# Assign issues to Copilot



# Create and review PRs

# Extend using MCP

# Coding agent features demo part one

# Best practices

CODING AGENT

## Well-scoped issues

Making sure your issues are well-scoped

## Choosing tasks

Choosing the right type of tasks to give to Copilot

## Use comments

Using comments to iterate on a pull request

## Custom instructions

Adding custom instructions to your repository

## Extend capabilities

Using the Model Context Protocol (MCP)

## Pre-install dependencies

Pre-installing dependencies in GitHub Copilot's environment

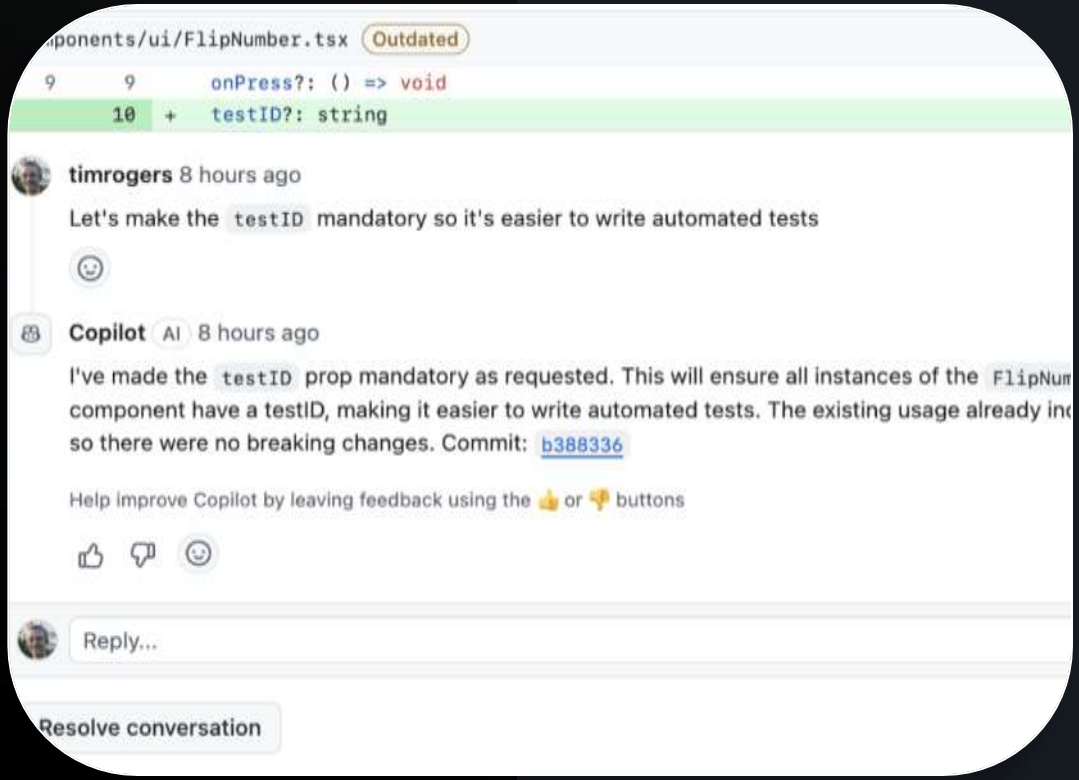
# Issues are well-scoped

- A clear description of the problem to be solved or the work required.
- Complete acceptance criteria on what a good solution looks like (for example, should there be unit tests?).
- Directions about which files need to be changed.

# Choosing the right type of tasks

- Complex and broadly scoped tasks
- Sensitive and critical tasks
- Ambiguous tasks
- Learning tasks

# Comments on a pull request



# Custom instructions

```
 is a Go based repository with a Ruby client for certain API endpoints. It i
imarily responsible for ingesting metered usage for GitHub and recording that
usage. Please follow these guidelines when contributing:

## Code Standards

### Required Before Each Commit
- Run `make fmt` before committing any changes to ensure proper code formatting
- This will run gofmt on all Go files to maintain consistent style

### Development Flow
- Build: `make build`
- Test: `make test`
- Full CI check: `make ci` (includes build, fmt, lint, test)

## Repository Structure
- `cmd/`: Main service entry points and executables
- `internal/`: Logic related to interactions with other GitHub services
- `lib/`: Core Go packages for billing logic
- `admin/`: Admin interface components
- `config/`: Configuration files and templates
- `docs/`: Documentation
- `proto/`: Protocol buffer definitions. Run `make proto` after making updates
here.
- `ruby/`: Ruby implementation components. Updates to this folder should include
incrementing this version file using semantic versioning: `ruby/lib/billing-
platform/version.rb`
- `testing/`: Test helpers and fixtures

## Key Guidelines
1. Follow Go best practices and idiomatic patterns
2. Maintain existing code structure and organization
3. Use dependency injection patterns where appropriate
4. Write unit tests for new functionality. Use table-driven unit tests when
possible.
   Document public APIs and complex logic. Suggest changes to the `docs/` folder
   appropriate
```

# Extend coding agent with MCP

MCP provides a standardized way to connect AI models to different data sources and tools, enabling them to work together more effectively. Below are some best practices

## Using MCP

### Enabling third-party

MCP servers for use may impact the performance of the agent and the quality of the outputs.

### Check write access

Update the tools field in the MCP configuration with only the necessary tooling.

### Carefully review

Carefully review the configured MCP servers prior to saving the configuration to ensure the correct servers are configured for use.

# Pre-install dependencies in GitHub Copilot environment

- Powered by GitHub Actions, Copilot has access to its own ephemeral development environment
- Configure a *copilot-setup-steps.yml* file to pre-install these dependencies

# Managing Copilot coding agent

## Enablement

The ability to use Copilot coding agent is controlled by policy settings defined at the organization level.

## Opting repos out

Opt out repositories and prevent Copilot coding agent from being used in those repositories. Members who have ability:

- Enterprise administrators and organization owners (for organization-owned repositories) and
- Users (for user-owned repositories)

# Coding agent best practices demo part two

# Questions?

# Thank you